

Econ 131
Fall 2019
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Final

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Student Name:

Student ID:

Exam Instructions

- **Closed book/notes exam.** No computer, calculator, or any electronic device allowed.
- **No phones.** Turn them off and put them in your bag.
- **Explanation should be written using pens.** No pencils, except for graphs.
- **You must submit your solutions using the exam packet provided.** If you need more room to write your answers or need to re-draw a graph use the extra pages at the end. Make sure to note it clearly and accurately if your solutions continue on a different page.
- **Do not write your solutions on pages that say “Do not write on this page”.** Answers written on these pages will not be graded.
- **When time is called, STOP** writing, immediately **CLOSE** your exam packet and hold it up until it is collected by one of the GSIs.

Do NOT open this test until instructed to do so.

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1. True/False/Uncertain with Explanation (50 points, 5 points per question.)

Explain your answer fully based on what was discussed in class, since all the credit is based on the explanation. Your grade depends entirely on the substance of your justification, not on whether you are correct in writing “True,” “False,” or “Uncertain.” Note that it is possible to answer each question for full credit with three sentences or fewer, and answers longer than ten lines long will not be graded.

- (a) In states with generous welfare benefits, people do not work very much. That means that generous welfare benefits cause people to work less.

- (b) Suppose that the government subsidizes coffee. That will create deadweight loss.

- (c) Suppose that instead of a coffee subsidy, there is a coffee tax of \$0.50 per cup of coffee. Consider two ways of applying the tax. In Scenario A, the tax is assessed on coffee producers and appears in the posted sale price in stores. In Scenario B, the tax is assessed on coffee consumers and does not appear in the posted sale price in stores. The number of coffee cups purchased under Scenario A will be larger than the number of coffee cups purchased under Scenario B.
- (d) The Earned Income Tax Credit is economically desirable because extensive-margin elasticities of labor supply (i.e. participation responses) are small.

(e) A public good can be thought of as a private good with widespread positive consumption externalities.

(f) The United States uses local property taxes rather than local per-person lump-sum taxes (sometimes called “poll taxes”). This entirely prevents “Tiebout sorting”: rich people could try to all move into big houses in towns with high property tax rates that fund great schools, but then lower earners could always move into small cheap homes in the town, pay low property taxes since their houses are cheap, and bankrupt the town.

(g) There is good evidence that when borrowing costs and tuition costs fall, more people attend college.

(h) Increasing the number of Americans who have health insurance will likely have large measurable health benefits, while making health insurance coverage more generous among those who already have insurance will likely have smaller measurable health benefits.

- (i) There is no particular reason that the Affordable Care Act (“ObamaCare”) needed to include all three of the following components, as opposed to just one or two: “community rating” rules (i.e. forbidding price discrimination based on preexisting conditions), an insurance mandate, and health insurance subsidies.

- (j) The evidence suggests one of the benefits of longer unemployment insurance durations is that people find jobs that pay more and last longer.

2. Public Goods, Externalities (15 points)

The residents of Smallville want the city government to build a new park. However, there is debate over how big to make the park. Smallville has 200 residents: 100 who live close to where the park will be built (and will benefit a lot from the new park), and 100 who live farther away (and will benefit less). The marginal benefits of the new park for each resident are given by the below equations, where A represents the size of the park in acres:

$$\text{For Each Close Resident: } MB_C = 115 - A$$

$$\text{For Each Far Resident: } MB_F = 25 - 2A$$

- (a) If construction of the park has a monetary cost to the city of \$5,000 per acre, how many acres should the city build to achieve the socially optimal number of acres A ? (3 points)

- (b) Now assume that since construction is noisy, residents living close to the site of the new park will suffer from the construction according to the following formula:

$$\text{For Each Close Resident: } MC_C = 2A + 30$$

The far away residents do not suffer any extra cost. Assuming the construction cost is still \$5,000 per acre, what is the new socially optimal number of acres A ? (2 points)

- (c) Now we have to think about how to pay for the park. Assume that the cost will be split evenly among the residents, so that each individual must bear $\frac{1}{200}$ of the \$5,000 per acre monetary cost of construction. If the city government has a rule that it will only build as many acres as the minimum demanded by any resident, how many acres will it now build? (For example, if 100 people demand 2 acres, and 100 demand 3 acres, the city will build 2 acres—the minimum amount demanded by any single resident.) (3 points)

Hint: Don't forget to consider noise when calculating how many acres each resident wants.

Now, instead of making the residents pay, the city will implement a Pigouvian tax on the emissions of a local factory. The marginal cost of the factory to reduce its emissions is given by:

$$MC = 100Q$$

where Q is the quantity of the reduction. Each of the residents benefits from the reduction of emissions according to the following formulas:

$$\text{For Each Close Resident: } MB_C = 20 - \frac{1}{2}Q$$

$$\text{For Each Far Resident: } MB_F = 10 - \frac{3}{2}Q$$

You should assume that the factory enjoys no marginal benefit of reduction and the residents face no marginal costs of reduction.

(d) What is the socially optimal amount of emissions reduction? (Don't forget how many residents there are.) (2 points)

(e) (*Harder*) What is the deadweight loss from the externality? (2 points)

(f) What flat tax t per unit of emissions (i.e. an amount assessed per unit of emissions) will cause the factory to reduce emissions by the socially optimal amount? (2 points)

3. Capital Tax, Social Security (10 points)

Consider a 2 period model where individuals earn labor income w from working in period 1 and do not work in period 2 (retirement). Individuals choose how much to consume in each period. Savings in period 1 earn an interest rate r . Let C_1 denote consumption in period 1 and C_2 denote consumption in period 2. Suppose that individuals have a utility function given by:

$$U = \ln C_1 + \ln C_2$$

- (a) Assume that the government imposes a capital tax τ on the individual's income from saving. Set up the individual's lifetime utility maximization problem as a utility function subject to some budget constraint. Do not solve it—just set it up. (2 points)
- (b) What are the signs (or directions) of the income and substitution effects of the capital tax on the individual's optimal choice of C_1 and C_2 ? Can we say definitively whether either C_1 or C_2 will increase or decrease? (2 points)

	Income Effect	Substitution Effect	Definitive Direction?
C_1			
C_2			

(c) Now assume there are no taxes. Consider three models of utility:

- Model 1: $U = \ln C_1 + \delta \ln C_2$, where $\delta = 1$
- Model 2: $U = \ln C_1 + \delta \ln C_2$, where $\delta < 1$
- Model 3: $U = \ln C_1 + \delta \ln C_2$, where $\delta > 1$

If we think of period 1 as the working years and period 2 as retirement, which model most accurately reflects evidence about consumption presented in lecture, and in which direction will C_1 and C_2 change compared to an agent under the utility function you used above in (a) and (b)? (2 points)

Most Accurate Model ?

Direction of C_1 ?

Direction of C_2 ?

Now, let's say the government wants to force saving for retirement, and so it establishes a social security program where it taxes each individual's labor income w with a labor tax τ_L in period 1, and then subsequently uses the tax revenue it collects from each individual to subsidize that *same* individual's retirement consumption in period 2. Assume there are no other taxes.

(d) Is the social security program described here funded or unfunded? Why? (2 points)

(e) If the government implements this social security program, would you expect the program to crowd out more private saving if the people are on average more myopic or less myopic. Why? (2 points)

Keep answers short, above this line.

4. Business Tax (10 points)

For this question, assume the US tax code imposes a corporate income tax $\tau_c = .20$, a personal income tax of $\tau_p = .40$, and a dividend tax of $\tau_d = .20$. Assume there are no other taxes.

Emily is an entrepreneur who is starting a business. The gross profit of her business (i.e. not including capital expenditures) is given by $F(K) = 20K^{1/2}$ and her cost of capital is given by $r = 2$ per unit of capital K she chooses to invest.

- (a) Assume that Emily plans to distribute all of the business profits to herself immediately as they are earned. If Emily wants to maximize her after-tax income as the owner, should she set up the business as a C-Corporation or S-Corporation? (2 points)

- (b) Regardless of your answer for part (a), assume for parts (b) and (c) that Emily chooses to set up the company as an S-Corporation. If there are no deductions for capital expenditures, what is Emily's optimal choice of capital investment K ? (2 points)

- (c) Now, let's say that policymakers enact a *partial* deduction for business capital expenditures: businesses are now able to deduct 50% of their capital expenditures. Write out Emily's new profit maximization problem. Do not solve it—just set it up. (2 points)

- (d) Using the theory and the evidence discussed in lecture, circle the best choice of words from each pair below to complete the sentence: (2 points)

“To minimize investment distortions while maintaining or increasing tax revenue, economic theory recommends that we **NARROW** / **BROADEN** the business income tax base and **RAISE** / **LOWER** the tax rate on business income. Over recent years, US business tax reform proposals on both sides of the political spectrum have tended to **FOLLOW** / **DO THE OPPOSITE OF** what theory recommends.”

- (e) For this final part (e) only, assume now that Emily set up her company as a C-Corporation. Let's say that policymakers raise τ_d from 0.20 to 0.25. Assume that capital expenditures are not deductible. How would the New View and the Traditional View predict Emily's optimal K to change? (You don't need to explain the views or do any math, just say the predictions for how optimal K changes) (2 points)

Keep answers short, above this line.

5. Unemployment Insurance (15 points)

Assume for this problem that individuals have a utility function given by:

$$U = \sqrt{c}$$

Where c is consumption. When individuals are employed and working, they earn wage $w = 100$. However, if they lose their job, for the time that they are unemployed they earn $w = 0$. You should assume that individuals spend 100% of their earnings on consumption, aside from whatever they spend on insurance.

In this world, however, individuals do not have an accurate perception of their own risk of unemployment. Individuals *think* that the probability of being unemployed in any given period is $q_i = 0.1$. The *actual* probability, q_a , is higher.

- (a) What is the individuals' expected utility of being uninsured as they perceive it (i.e., based upon the assumption that they think their risk of unemployment is $q_i = 0.1$)? (2 points)

- (b) Private insurers enter the market and offer unemployment insurance to individuals, with premium p (that must be paid regardless of whether the individual is employed or unemployed) and payouts b (i.e. the benefits received when the individual is unemployed). What is the value of b that the individuals will want to buy? How can you know this quickly from the information given above? (2 points)

- (c) Let's say that private insurers estimate that the *actual* likelihood of unemployment is $q_a = 0.3$. Assuming this number is accurate, what price will the insurers need to charge to break even? (Assume that insurers know the b you calculated in part (b).) (1 point)

(d) Using individuals' perceptions of their own riskiness q_i , calculate whether the individuals will buy insurance at the market price you calculated in (b). (3 points)

(e) Is the behavior in (d) an example of adverse selection? Why or why not? (2 points)

Keep answers short, above this line.

- (f) Now assume that everyone knows their true chance of unemployment is $q_a = 0.3$ (that is, you should ignore q_i now). If the insurance industry became a monopoly and raised the premiums, what is the highest premium p the insurer could charge before the buyers drop out? (3 points)

- (g) **Deadweight Loss of a Payroll Tax.** Now assume that the government decides to provide unemployment insurance to all workers. To pay for the new program, the government imposes a new payroll tax t on employers (i.e. a tax paid by employers on the wages paid to employees).

If we assume that the equilibrium quantity of labor and wage before the new tax is imposed is given by Q_L and w , and that this is an undistorted equilibrium, and that the price elasticities on the labor supply side and labor demand side are given by $\varepsilon_S = 1$ and $\varepsilon_D = -0.5$, respectively, write down a formula for the expected deadweight loss when this new payroll tax t is imposed in terms of Q_L , w , and t . (2 points)

END OF EXAM

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